

Actors:

User

IT User

Use Case List:

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UC - 13: Export Chat Log

UC - 14: IT Exit Audit Mode

UC - 15: Start Server

UC - 16: Stop Server

Use Case ID: 0

Use Case Name: Create User

Relevant Requirements: N/A

Primary Actor: User/ITUser

Pre-Conditions: User account doesn't exist

Post-Conditions: New user account created

Main Scenario:

1. The actor opens the application and the log in screen is shown.
2. The actor enters username and password.
3. Client sends credentials to the CSM for account creation.
4. CSM validates the credentials checking if the user is not already created

Extensions:

1. If client creates an IT account, the user created is an IT user
2. If not, the user created is a User

Related Use Cases: UC-1

Use Case ID: 1

Use Case Name: Log In

Relevant Requirements: 3.1.1, 3.2.1-3.2.6, 3.5.5-3.5.8, 4.1.1

Primary Actor: User/IT User

Pre-Conditions: User account exists, client is running, server unreachable

Post-Conditions: User session established, user role determined, user routed to CM

Basic Flow or Main Scenario:

1. The actor opens the application and the log in screen is shown.
2. The actor enters username and password.
3. Client sends credentials to the CSM for authentication.
4. CSM validates the credentials and determines the role. (User or IT)
5. CSM returns authentication result.
6. Client displays the correct UI. (Client Module for User or IT Module for IT)

Extensions or Alternate Flows:

1. If User: show Communications interface.
2. If IT User: show IT interface.

Exceptions:

1. Invalid username/password: show failure message/prompt re-entry.
2. Server unreachable: show connection error/remain on login screen.

Related Use Cases: UC-2, UC-11

Use Case ID: 2

Use Case Name: Log Out

Relevant Requirements: 3.312-3.313, 3.5.13

Primary Actor: User/IT User

Pre-Conditions: Actor is logged in

Post-Conditions: Session terminated, user is returned to login menu, server marks user offline

Basic Flow or Main Scenario:

1. The actor selects Profile or account menu.
2. Actor selects Logout.
3. Client notifies CSM of logout.
4. CSM marks users as offline.
5. Client returns to the login menu.

Extensions or Alternate Flows: N/A

Exceptions:

1. Server unreachable: client still ends local session and returns to login.

Related Use Cases: UC-1

Use Case ID: 3

Use Case Name: Open Chat

Relevant Requirements: 3.3.14-3.3.19, 3.5.8, 3.5.11

Primary Actor: User

Pre-Conditions: User logged in, user is a participant in the chat

Post-Conditions: Messages displayed, offline/unread messages marked seen, client updates server status

Basic Flow or Main Scenario:

1. The user selects a chat from the chat list.
2. Client requests chat history and unread/offline messages have been seen.

Extensions or Alternate Flows:

1. While chat is open, if a new message arrives, client updates the view automatically.

Exceptions:

1. User is not authorized for chat: deny and return to chat list.

Related Use Cases: UC-4

Use Case ID: 4

Use Case Name: Send Message

Relevant Requirements: 3.3.20-3.3.22, 3.5.8-3.5.10

Primary Actor: User

Pre-Conditions: User logged in, chat open, user is a chat participant

Post-Conditions: Message stored, delivered to online chat participants, appears in chat history

Basic Flow or Main Scenario:

1. The user types a message and clicks Send.
2. Client attaches the current timestamp and sends a message to CSM.
3. CSM stores the message in file storage.
4. CSM delivers the message to online participants in the chat.
5. Client updates chat view to show the new message.

Extensions or Alternate Flows: N/A

Exceptions:

1. Server unreachable: show send failure

Related Use Cases: UC-3

Use Case ID: 5

Use Case Name: Create Private Chat

Relevant Requirements: 3.3.8-3.3.11

Primary Actor: User

Pre-Conditions: User logged in

Post-Conditions: New private chat created, chat appears in list

Basic Flow or Main Scenario:

1. User clicks New Chat.
2. The user enters exactly one username.
3. Client sends chat creation request to CSM.
4. CSM creates private chat and returns success.
5. Client updates chat list to include new chat.

Extensions or Alternate Flows: N/A

Exceptions:

1. Username invalid/not found: show error/do not create chat.
2. Chat already exists: server returns existing chat.

Related Use Cases: UC-4

Use Case ID: 6

Use Case Name: Create Group Chat

Relevant Requirements: 3.3.8-3.3.11

Primary Actor: User

Pre-Conditions: User logged in

Post-Conditions: Group chat created, creator recorded, chat appears in list

Basic Flow or Main Scenario:

1. User clicks New Chat.
2. User selects "Group Chat" (or multiple usernames).
3. Client sends group chat creation request to CSM.
4. CSM creates private chat and returns success.
5. Client updates chat list to include new chat.

Extensions or Alternate Flows: N/A

Exceptions:

1. One or more usernames are invalid: show error/do not create chat.

Related Use Cases: UC-7, UC-8, UC-9

Use Case ID: 7

Use Case Name: Add User to Group Chat

Relevant Requirements: 3.3.24-3.3.25

Primary Actor: User (Group Chat Creator)

Pre-Conditions: User logged in, group chat open, user is creator

Post-Conditions: New member is added, member sees chat in list

Basic Flow or Main Scenario:

1. Creator clicks invite.
2. Creator enters username(s).
3. Client sends invite requests to CSM.
4. CSM validates and adds user(s) to the group.
5. CSM updates chat lists for users in the group.

Extensions or Alternate Flows: N/A

Exceptions:

1. Non-creator attempts invite: deny.
2. Invalid username: error.

Related Use Cases: UC-6

Use Case ID: 8

Use Case Name: Remove User from Group Chat

Relevant Requirements: 3.3.26

Primary Actor: User (Group Chat Creator)

Pre-Conditions: User logged in, group chat open, user is creator

Post-Conditions: Selected member is removed, member no longer sees chat in list

Basic Flow or Main Scenario:

1. The creator clicks Kick.
2. Creator selects a member to remove.
3. Client sends kick request to CSM.
4. CSM removes members from group chat.
5. CSM updates chat lists accordingly.

Extensions or Alternate Flows: N/A

Exceptions:

1. Attempt to kick creator: deny.
2. Non-creator attempts to kick: deny.

Related Use Cases: UC-6, UC-7

Use Case ID: 9 (optional, time constraint)

Use Case Name: Delete Group Chat

Relevant Requirements: 3.3.27-3.3.28

Primary Actor: User (Group Chat Creator)

Pre-Conditions: User logged in, group chat exists, user is creator

Post-Conditions: Chat removed from users' views, messages remain stored but hidden

Basic Flow or Main Scenario:

1. Creator clicks Delete Group Chat.
2. System prompts for confirmation.
3. The creator confirms.
4. Client sends delete request to CSM.
5. CSM marks a group chat as deleted.
6. Messages are remain stores in history

Extensions or Alternate Flows: N/A

Exceptions:

1. Non-creator attempts to delete: deny.

Related Use Cases: UC-6

Use Case ID: 10

Use Case Name: IT Enter Audit Mode

Relevant Requirements: 3.4.2-3.4.7

Primary Actor: IT User

Pre-Conditions: IT user logged in

Post-Conditions: Audit mode active for selected user, selected user's chat list is displayed

Basic Flow or Main Scenario:

1. The IT user selects "Change View".
2. System displays User List.
3. IT user searches/selects a user.
4. The system enters Audit Mode for that user and displays their chat list.

Extensions or Alternate Flows: N/A

Exceptions:

1. Selected user not found: show error.

Related Use Cases: UC-11, UC-12

Use Case ID: 11

Use Case Name: IT Select User

Relevant Requirements: 3.4.8-3.4.11

Primary Actor: IT User

Pre-Conditions: IT user in Audit Mode

Post-Conditions: The system display the chats of selected user

Basic Flow or Main Scenario:

1. IT enter Audit mode
2. From a scrollable list of names, IT users click on a name
3. Selected user data will be displayed

Extensions or Alternate Flows: N/A

Exceptions:

1. IT user attempts to send/edit/delete: deny.

Related Use Cases: UC-12

Use Case ID: 12

Use Case Name: IT View Chats

Relevant Requirements: 3.4.8-3.4.11

Primary Actor: IT User

Pre-Conditions: IT user in Audit Mode, selected user has chats

Post-Conditions: Conversation history is displayed, no changes allowed.

Basic Flow or Main Scenario:

4. The IT user opens a chat in the selected user's chat list.
5. The system displays full message history in a scrollable view.
6. The system indicates audit mode visually.
7. The system prevents sending/editing/deleting actions.

Extensions or Alternate Flows: N/A

Exceptions:

2. IT user attempts to send/edit/delete: deny.

Related Use Cases: UC-13

Use Case ID: 13

Use Case Name: Export Chat Log

Relevant Requirements: 3.4.12

Primary Actor: IT User

Pre-Conditions: IT user in Audit Mode, chat open

Post-Conditions: Export file saved locally to IT user's machine

Basic Flow or Main Scenario:

1. IT user selects Export Chat.
2. System prompts for local save location.
3. The system writes the displayed chat log to a local file.
4. The system confirms export success.

Extensions or Alternate Flows: N/A

Exceptions:

1. File writes fails: show error.

Related Use Cases: UC-12

Use Case ID: 14

Use Case Name: IT Exit Audit Mode

Relevant Requirements: 3.4.13

Primary Actor: IT User

Pre-Conditions: Audit mode active

Post-Conditions: User Routed to the CM

Basic Flow or Main Scenario:

1. IT user selects exit view
2. System displays the CM UI

Extensions or Alternate Flows: N/A

Exceptions: N/A

Related Use Cases: UC-2, UC-10, UC-12, UC-13

Use Case ID: 15

Use Case Name: Start Server

Relevant Requirements: N/A

Primary Actor: Server Host

Pre-Conditions: Server offline

Post-Conditions: Server online

Basic Flow or Main Scenario:

1. Server is ran by the server host
2. The local data is loaded
3. Server is online to accept client connections

Extensions or Alternate Flows: N/A

Exceptions: N/A

Related Use Cases: UC-16

Use Case ID: 16

Use Case Name: Start Server

Relevant Requirements: N/A

Primary Actor: Server Host

Pre-Conditions: Server online

Post-Conditions: Server offline

Basic Flow or Main Scenario:

1. Server is stopped by the Server Host
2. The local data is saved
3. Server goes offline

Extensions or Alternate Flows: N/A

Exceptions: N/A

Related Use Cases: UC-16